

IONQ2026

1. Quantum Entanglement Distillation Game

One sentence: quantum networking game set on a simulated quantum network

- Start from a single node
 - Aim to expand a connected subgraph by claiming edges thr entanglement distillation
 - **while competing against other players under strict resource constraints**
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2. Objective

- Build a connected quantum network subgraph
 - Maximize **utility qubits (score)** obtained from network nodes
 - Manage a limited budget of Bell pairs while competing with others
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3. Game Environment

- **Vertices (Nodes)**
Each node provides:
 - Utility qubits (points)
 - Bonus Bell pairs (additional resources)
 - Limited capacity (only a fixed number of players can earn rewards)
 - **Edges (Links)**
 - Represent noisy entanglement connections between nodes.
 - An edge can only be claimed by successfully distilling a high-fidelity Bell pair
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4. Core Gameplay

- Players design and submit **quantum circuits** that perform **entanglement distillation**.
- Multiple noisy Bell pairs are transformed into a higher-fidelity Bell pair.

- If the resulting fidelity exceeds the edge's threshold, the edge is claimed.
 - Successful claims consume Bell pair resources; failed attempts do not.
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5. Constraints

- Bell pairs are limited and must be managed carefully.
- Only local quantum operations and classical communication (LOCC) are allowed.
- All players compete simultaneously for node rewards